

Reproducing MoE-CP: Adaptive Conformal Prediction via Mixture-of-Experts Gating Similarity

October 2026

Abstract

We reimplement MoE-CP and its five baselines and repeat the paper’s four experiments with every setting the paper states. MoE-CP’s median interval length is within 2.1% of the reported value in all four experiments, at the nominal 90% coverage, and it is the shortest method in each, as reported. The baselines agree within about 6%, except PCP on synthetic data, which is 17% shorter and less conservative here.

1 Algorithm

Let $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{cal}} = \{(X_i, Y_i)\}_{i=1}^n$ be i.i.d. samples and X_{n+1} a test point. MoE-CP [1] returns an interval $\hat{C}(X_{n+1})$ with $\mathbb{P}\{Y_{n+1} \in \hat{C}(X_{n+1})\} \geq 1 - \alpha$ whose width adapts to the latent regime of X_{n+1} .

Predictor. A mixture of experts with K expert networks μ_k and a gating network with logits ℓ_k is trained jointly on $\mathcal{D}_{\text{train}}$ with squared loss:

$$\pi_k(x) = \frac{\exp \ell_k(x)}{\sum_{j=1}^K \exp \ell_j(x)}, \quad \hat{\mu}(x) = \sum_{k=1}^K \pi_k(x) \mu_k(x). \quad (1)$$

The gating vector $\pi(x) = (\pi_1(x), \dots, \pi_K(x))$ serves as a soft domain assignment.

Calibration. Scores are absolute residuals, $S_i = |Y_i - \hat{\mu}(X_i)|$. The test gating vector is randomized,

$$\tilde{L} \sim \text{Multinomial}(\tau, \pi(X_{n+1})), \quad \tilde{\pi} = \tilde{L}/\tau, \quad (2)$$

and each calibration point, together with the test point itself, is weighted by its gating similarity to $\tilde{\pi}$:

$$w_i = \exp[-\tau D(\tilde{\pi}, \pi(X_i))], \quad \tilde{w}_i = \frac{w_i}{\sum_{j=1}^{n+1} w_j}, \quad i = 1, \dots, n+1, \quad (3)$$

with $D = \text{KL}$ throughout. The interval is

$$\hat{C}(X_{n+1}) = \left\{ y : |y - \hat{\mu}(X_{n+1})| \leq Q_{1-\alpha} \left(\sum_{i=1}^n \tilde{w}_i \delta_{S_i} + \tilde{w}_{n+1} \delta_{+\infty} \right) \right\}. \quad (4)$$

For the KL divergence (equivalently cross-entropy), coverage is at least $1 - \alpha$ conditionally on $\tilde{\pi}$ [1, Thm. 1]. As $\tau \rightarrow 0$ the method reduces to split conformal prediction; when $\tilde{w}_{n+1} > \alpha$ the interval is the whole line. Algorithm 1 lists the steps as implemented.

Algorithm 1 MoE-CP (absolute-residual score, KL divergence)

Require: $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{cal}}$, test point X_{n+1} , experts K , temperature τ , level α

- 1: Fit $\{\mu_k, \ell_k\}_{k=1}^K$ on $\mathcal{D}_{\text{train}}$ by minimizing the squared error of $\hat{\mu}$ in (1)
 - 2: **for** $i = 1, \dots, n$ **do** $\triangleright$ calibration set
 - 3: $S_i \leftarrow |Y_i - \hat{\mu}(X_i)|$; store $\pi(X_i)$
 - 4: **end for**
 - 5: Draw $\tilde{L} \sim \text{Multinomial}(\tau, \pi(X_{n+1}))$ and set $\tilde{\pi} \leftarrow \tilde{L}/\tau$
 - 6: $w_i \leftarrow \exp[-\tau \text{KL}(\tilde{\pi} \parallel \pi(X_i))]$ for $i = 1, \dots, n+1$; $\tilde{w}_i \leftarrow w_i / \sum_j w_j$
 - 7: $Q \leftarrow \inf\{s : \sum_{i \leq n: S_i \leq s} \tilde{w}_i \geq 1 - \alpha\}$ $\triangleright +\infty$ if never reached
 - 8: **return** $[\hat{\mu}(X_{n+1}) - Q, \hat{\mu}(X_{n+1}) + Q]$
-

2 Datasets

Synthetic. Three latent domains share the covariate but differ in the regression function:

$$X \sim \text{Uniform}[-3, 3], \quad \varepsilon \sim N(0, 1), \quad Y = \begin{cases} X + \varepsilon & \text{with probability } p_1, \\ X^2 + \varepsilon & \text{with probability } p_2, \\ X^3 + \varepsilon & \text{with probability } p_3. \end{cases} \quad (5)$$

Setting 1 is unbalanced, $(p_1, p_2, p_3) = (0.2, 0.3, 0.5)$; Setting 2 is balanced, $p_k = 1/3$. The domain label is not observed. Each repetition draws three independent samples of size 500 for training, calibration and testing.

Bike sharing. UCI Bike Sharing dataset¹, file `hour.csv`: 17,379 hourly records. The response is the total rental count `cnt`. The 13 features are all remaining columns except the date string and the two components of the response (`casual`, `registered`): `instant`, `season`, `yr`, `mnth`, `hr`, `holiday`, `weekday`, `workingday`, `weathersit`, `temp`, `atemp`, `hum`, `windspeed`, with categorical variables kept as the integer codes shipped in the file.

Temperature. UCI bias-correction dataset for LDAPS temperature forecasts over Seoul², summers 2013–2017: 7,752 records, 7,590 after removing those with missing values. The response is the next-day minimum temperature `Next_Tmin`; `station`, `Date` and `Next_Tmax` are dropped, leaving 21 predictors.

Splitting. For the real datasets each repetition draws 4,500 records uniformly without replacement and splits them into disjoint training, calibration and test sets of 1,500 each. Features and response are standardized with the mean and standard deviation of the training set, so interval lengths are in units of the response’s standard deviation. Synthetic data are used in raw units. Table 1 summarizes the protocol.

3 Parameters

Table 2 lists every setting the paper states (Section 4 and Appendix A.4.1); the implementation uses each of them unchanged.

¹<https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset>

²<https://archive.ics.uci.edu/dataset/514/bias+correction+of+numerical+prediction+model+temperature+forecast>

Table 1: Data protocol. Every experiment is repeated 50 times with a fresh draw.

Dataset	Records	Features	Response	Train / cal / test	Standardized
Synthetic (Settings 1, 2)	generated	1	Y	500 / 500 / 500	no
Bike sharing	17,379	13	cnt	1500 / 1500 / 1500	yes
Temperature	7,590	21	Next_Tmin	1500 / 1500 / 1500	yes

Table 2: MoE-CP settings. All values are those stated in the paper.

Setting	Synthetic	Real data
Experts K	3	2
Hidden layers, experts and gate	(32, 32, 32)	(64, 64, 64)
Activation, dropout rate	ReLU, 0.1	ReLU, 0.1
Batch size, learning rate	64, 10^{-4}	64, 10^{-4}
Training epochs	600	2000
Temperature τ	150	100
Divergence D	KL	KL
Miscoverage α	0.1	0.1
Training / calibration / test size	500 each	1500 each
Repetitions	50	50

Baselines. CQR [2] and CHR [3] use a quantile random forest (100 trees, minimum leaf size 50, quantile grid $0.01, \dots, 0.99$). SCP+CC [4], RLCP [5] and PCP [6] use a random forest with scikit-learn’s default settings, as in the paper. RLCP’s bandwidth gives an effective sample size of 100. SCP+CC uses the linear basis $(1, x)$ on the signed residual, with randomized bounds at levels $\alpha/2$ and $1 - \alpha/2$. PCP keeps its reference defaults (20 folds; 9 residual levels between the 10th and 90th percentiles).

Choices the paper leaves open.

- *Optimizer.* The paper says “mini-batch gradient descent”; we use Adam.
- *Stopping.* The paper mentions “an early stopping ceiling on epochs”. We train for the full number of epochs on the whole training set; the paper’s runtimes, linear in sample size and equal across datasets, indicate the same.
- *Dropout.* Applied after every hidden layer.
- *Standardization.* Training-set statistics, recomputed in each repetition.
- *Bike features.* The paper gives the count (13) but not the list; it also reports 17,389 records where the file has 17,379.
- *Temperature records.* The paper reports 7,750 and does not mention missing values; we drop the 162 incomplete records.
- *Quantile forest.* Leaf size and tree count are not given; ours follow the CHR authors’ experiments.
- *Worst-slice coverage.* Slabs holding at least 10% of the points, 1000 random directions, searched on 20% of the test set and evaluated on the other 80%.
- *Infinite intervals* (MoE-CP, RLCP, PCP, SCP+CC) count as twice the largest absolute test residual, the convention of the baselines’ reference code.

4 Results

The paper reports results only as boxplots. Its numbers below are the box medians, read from the vector graphics of its Figures 2, 5 and 6. Ours are medians over the 50 repetitions.

Table 3: Synthetic data: median marginal coverage and interval length. Shortest length in bold.

Method	Setting 1 (unbalanced)				Setting 2 (balanced)			
	Coverage		Length		Coverage		Length	
	Paper	Ours	Paper	Ours	Paper	Ours	Paper	Ours
CQR	0.897	0.897	13.32	13.73	0.906	0.899	12.67	13.01
CHR	0.898	0.894	12.82	13.06	0.908	0.897	11.91	12.17
SCP+CC	0.902	0.900	22.46	21.20	0.897	0.894	19.36	19.42
RLCP	0.909	0.907	16.79	16.25	0.910	0.908	16.00	15.87
PCP	0.944	0.923	21.12	17.44	0.944	0.926	20.64	17.00
MoE-CP	0.911	0.907	11.74	11.84	0.911	0.906	11.91	11.88

Table 4: Real data: median marginal coverage, interval length (standardized units) and worst-slice coverage (WSC). Shortest length in bold.

Method	Bike sharing						Temperature					
	Coverage		Length		WSC		Coverage		Length		WSC	
	Paper	Ours	Paper	Ours	Paper	Ours	Paper	Ours	Paper	Ours	Paper	Ours
CQR	0.906	0.903	1.785	1.860	0.883	0.879	0.902	0.901	1.447	1.489	0.885	0.886
CHR	0.907	0.898	1.610	1.690	0.892	0.891	0.905	0.899	1.439	1.476	0.892	0.892
SCP+CC	0.902	0.899	1.116	1.072	0.884	0.895	0.902	0.901	1.189	1.196	0.891	0.889
RLCP	0.920	0.916	1.588	1.555	0.901	0.905	0.928	0.925	1.852	1.949	0.917	0.921
PCP	0.920	0.911	1.302	1.227	0.910	0.906	0.920	0.913	1.352	1.361	0.915	0.902
MoE-CP	0.906	0.900	1.077	1.054	0.885	0.855	0.903	0.902	1.067	1.073	0.903	0.885

MoE-CP. The reported behaviour is reproduced. Median length differs from the paper’s by +0.8% and −0.2% on the two synthetic settings, −2.1% on Bike sharing and +0.6% on Temperature, with coverage between 0.900 and 0.907. MoE-CP has the shortest median length in every experiment and is the shortest method in 90%, 60%, 64% and 100% of the repetitions, respectively. Its worst-slice coverage is lower than reported on both real datasets (0.855 against 0.885, and 0.885 against 0.903), though within the interquartile range of the paper’s boxes.

Baselines. The methods rank by length exactly as in the paper on Setting 1, Bike sharing and Temperature; on Setting 2 only PCP and SCP+CC exchange places. CQR and CHR are 2–5% longer than reported, and SCP+CC and RLCP are within 6%; we could not attribute the CQR and CHR gap to the forest settings or the feature set. PCP matches on Temperature but is less conservative on synthetic data (coverage 0.92 against 0.94, length 17% shorter). Our PCP corrects three defects of its reference code, which may explain this; we have not verified it.

Sensitivity to the open choices. Two of the choices in Section 3 matter. Training the mixture of experts with early stopping on a 10% held-out part of the training set (patience 100), and using day-of-month in place of `instant` as the thirteenth Bike feature, lengthened MoE-CP’s intervals

to 12.28 on Setting 1, 1.375 on Bike sharing and 1.127 on Temperature, that is 5%, 28% and 6% above the reported values.

Figures. Figures 1–4 are produced by the code and correspond to the paper’s Figures 2–5. Figures 2 and 3 show a single repetition of Setting 1. In it, the local coverage of SCP+CC falls to about 0.75 at both ends of the covariate range, where every other method stays at 0.85 or above; MoE-CP ranges from 0.87 to 1.00. MoE-CP, CQR and CHR widen towards the ends of the range, where the three domains separate, whereas SCP+CC keeps a nearly constant width.

References

- [1] J. Kong, W. Tang, D. Kong, L. Kong, G. Yang, B. Jiang. Adaptive conformal prediction via mixture-of-experts gating similarity. *ICLR*, 2026.
- [2] Y. Romano, E. Patterson, E. Candès. Conformalized quantile regression. *NeurIPS*, 2019.
- [3] M. Sesia, Y. Romano. Conformal prediction using conditional histograms. *NeurIPS*, 2021.
- [4] I. Gibbs, J. Cherian, E. Candès. Conformal prediction with conditional guarantees. *JRSS-B*, 2025.
- [5] R. Hore, R. F. Barber. Conformal prediction with local weights: randomization enables robust guarantees. *JRSS-B*, 2025.
- [6] Y. Zhang, E. Candès. Posterior conformal prediction. *arXiv:2409.19712*, 2024.

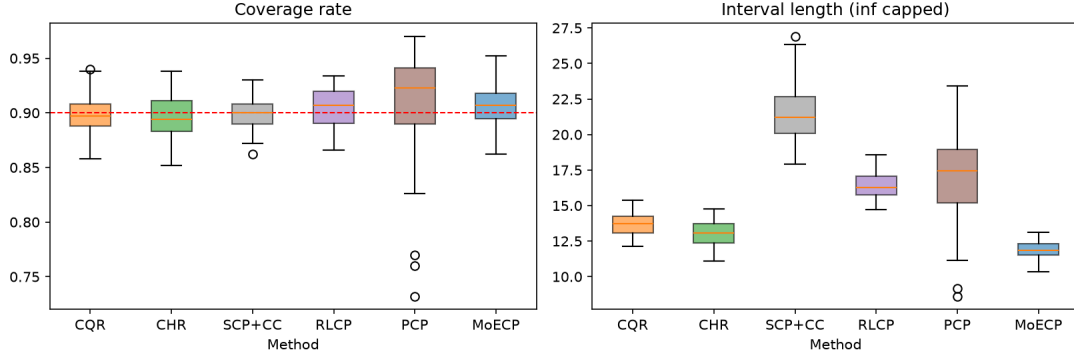


Figure 1: Synthetic data, Setting 1: marginal coverage and interval length over 50 repetitions (paper's Fig. 2).

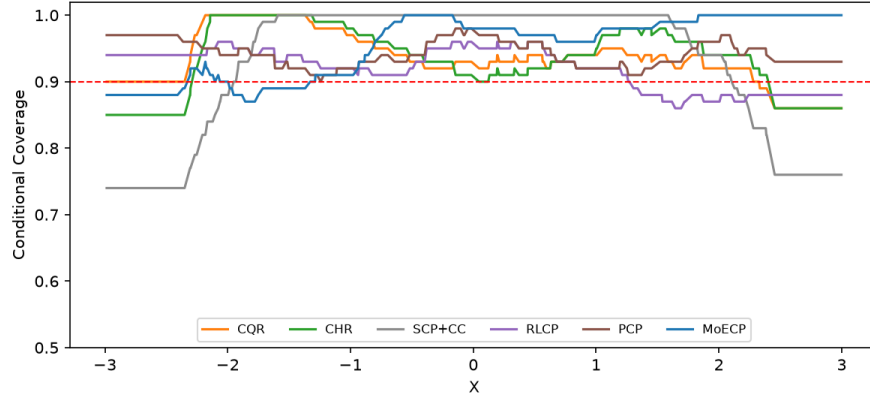


Figure 2: Synthetic data, Setting 1, one repetition: local coverage, averaged over each test point's 100 nearest neighbours (paper's Fig. 3).

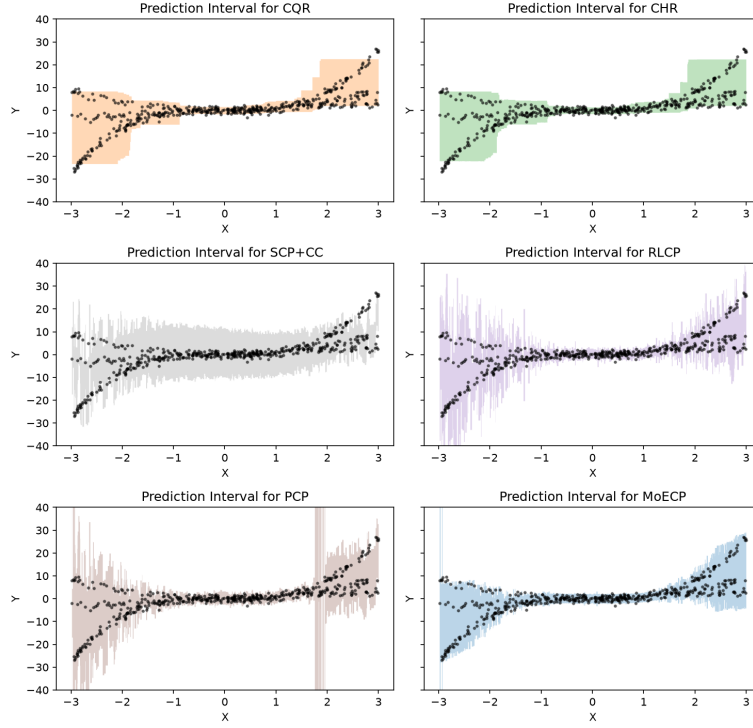


Figure 3: Synthetic data, Setting 1, one repetition: prediction intervals and test points. Infinite intervals are drawn to the edge of the frame (paper’s Fig. 4).

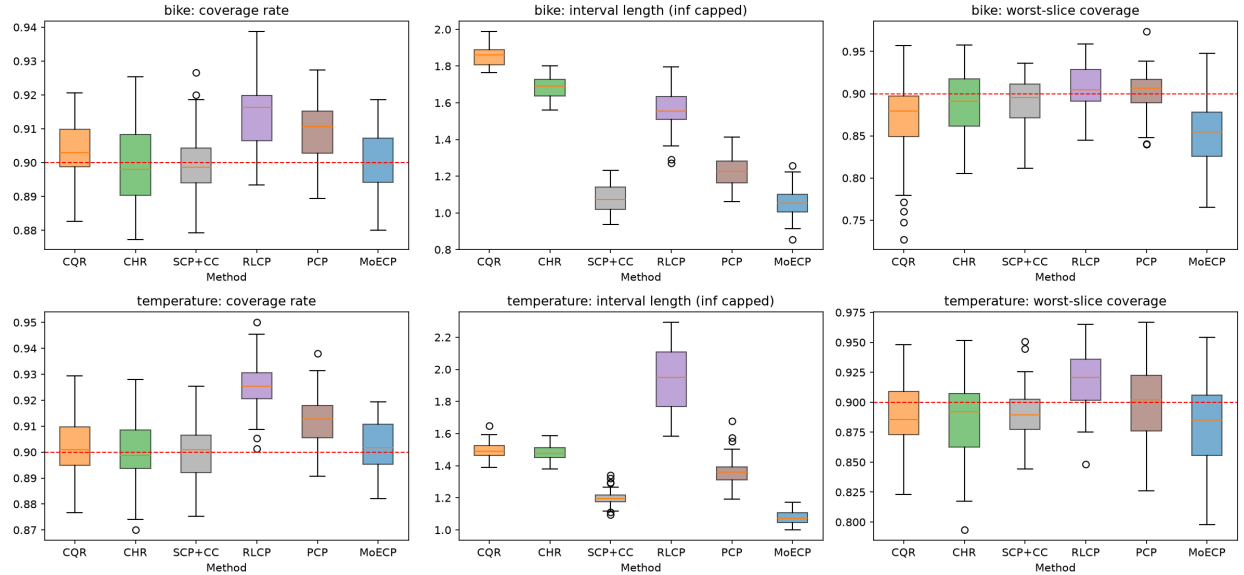


Figure 4: Real data over 50 repetitions: marginal coverage, interval length and worst-slice coverage for Bike sharing (top) and Temperature (bottom) (paper’s Fig. 5).